

Software Engineering LAB-01

Problem Statement #42: Internal Microservice Catalog & Health Portal

Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-001	Functional	The system shall ping microservice health endpoints at 30-second intervals and compute rolling 24-hour service availability percentages.	High	Pass: Downtime is recorded and an alert is dispatched after 3 consecutive failed health probes. Fail: An unreachable service is marked healthy.	Ensures continuous monitoring and accurate measurement of service availability.
FR-002	Functional	The system shall allow authorized users to view the catalog of registered microservices.	High	Pass: When an authorized user opens the service catalog, registered microservices and their relevant information are displayed. Fail: Registered services are missing from the catalog.	Provides a centralized view of available microservices.
FR-003	Functional	The system shall display dependencies between registered microservices using an interactive dependency graph.	High	Pass: Selecting a microservice displays its connected dependencies in the graph. Fail: Known dependencies are not displayed.	Helps users understand relationships and dependencies between services.
FR-004	Functional	The system shall provide aggregated API documentation for registered microservices.	Medium	Pass: When an authorized user selects a registered microservice, its available API documentation is displayed. Fail: API documentation cannot be accessed.	Provides a centralized location for microservice API information.
FR-005	Functional	The system shall record downtime events for microservices that fail the configured health-check condition..	High	Pass: When a microservice fails 3 consecutive health probes, a downtime event is recorded with the service name and failure time. Fail: No downtime event is recorded after the configured failure condition is reached.	Provides a history of service failures for monitoring and troubleshooting.

NFR001	Nonfunctional	The service catalog dependency graph viewer must render interactive node architectures containing up to 200 services smoothly.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load. Fail: The dependency graph does not meet the required performance or security standards.	Ensures the dependency graph remains usable for large service architectures.
NFR002	Nonfunctional	The system shall restrict access to the internal microservice catalog and API documentation to authenticated and authorized users.	High	Pass: An authenticated and authorized user can access the catalog and API documentation, while an unauthorized user is denied access. Fail: An unauthorized user can access the catalog or API documentation.	Protects internal microservice and API information from unauthorized access.